

Question:

A baby is born after the 32nd gestational week by cesarean delivery. The mother suffered from gestational diabetes; however, she had no other pregnancy-related diseases and was otherwise healthy. The baby has a blood pressure of 100/58 mm Hg, heart rate of 104/min, and oxygen saturation of 88%. The child has tachypnea, subcostal and intercostal retractions, nasal flaring, and cyanosis. The cyanosis is responding well to initial administration of oxygen. The nasogastric tube was positioned without problems. Which of the following is the most likely diagnosis?

Answer Choices:

- A. Neonatal respiratory distress syndrome (NRDS)
- B. Tracheoesophageal fistula
- C. Pneumonia
- D. Congenital heart anomaly with right-to-left shunt

Reasoning + Answer:

<think>
**Finding reasoning paths : **

- 1 . Tachypnea -> Respiratory distress -> Neonatal respiratory distress syndrome (NRDS)
- 2 . Subcostal and intercostal retractions -> Respiratory distress -> Neonatal respiratory distress syndrome (NRDS)
- 3 . Cyanosis -> Hypoxemia -> Neonatal respiratory distress syndrome (NRDS)
- 4 . Nasal flaring -> Respiratory distress -> Neonatal respiratory distress syndrome (NRDS)

**Reasoning Process : **

- 1 . **Tachypnea:** The baby exhibits tachypnea , which is an increased respiratory rate . This is often a sign of respiratory distress , where the body is trying to compensate for inadequate oxygenation or ventilation . Tachypnea is commonly seen in neonatal respiratory distress syndrome (NRDS) due to surfactant deficiency , which can lead to respiratory distress .
- 2 . **Subcostal and intercostal retractions:** These retractions indicate difficulty in breathing , which is typical of respiratory distress . In premature infants , this can be due to NRDS , where the lack of surfactant causes alveolar collapse and increased work of breathing .
- 3 . **Cyanosis:** Cyanosis is the bluish discoloration of the skin due to inadequate oxygenation . This can occur in NRDS because of insufficient oxygen reaching the tissues , leading to hypoxemia .
- 4 . **Nasal flaring:** Nasal flaring is another sign of respiratory distress , where the infant tries to increase oxygen intake by flaring the nostrils . This is often seen in conditions like NRDS , where the respiratory system is struggling to maintain adequate oxygenation .
- 5 . **Initial oxygen response:** The cyanosis is responding well to oxygen therapy suggests that the primary issue is related to oxygenation , which is consistent with NRDS , where supplemental oxygen can improve oxygen saturation temporarily until more definitive treatment can be provided .

**Conclusion : **

Based on the symptoms of tachypnea , subcostal and intercostal retractions , cyanosis , and nasal flaring , the most likely diagnosis for the baby is neonatal respiratory distress syndrome (NRDS). This condition is common in preterm infants due to surfactant deficiency , which matches the gestational age of the baby (born after the 32 nd gestational week , suggesting prematurity).

</think>
<answer>
Neonatal respiratory distress syndrome (NRDS)
</answer>